

Solution Requirements

Date	02 October 2026
Team ID	SWTID-2026-7991
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can register/sign in and maintain an authenticated session before using personalized learning features.	High
2	Authorization	Authenticated students can access their own learning features and generated results; access is limited to permitted functions.	High
3	External interfaces	Connect to the Google Gemini API to generate answers, explanations, quizzes, summaries and learning recommendations.	High

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4	Transactions processing	Accept a student request, validate the input, send the required prompt to Gemini, receive the response and display it in the learning interface.	High
5	Reporting	Provide learning history and quiz results so students can review generated content and self-assessment outcomes.	Medium
6	Business rules, etc.	Q&A should be relevant to the student query; explanations should be simplified; quizzes should contain exactly 3 MCQs with 4 options, answers and explanations; learning paths should progress from beginner to advanced.	High
7	Compliance to laws or regulations	Protect user information and application credentials; do not expose API keys or unnecessary personal data through the application.	Medium
8	Other	Provide dedicated functions for Q&A, concept explanation, quiz generation, educational-text summarization and personalized learning recommendations.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Requests should be processed promptly and the learning response should appear without unnecessary delay.	Typical request response within 5 seconds under normal load.
2	Scalability	The backend should support additional students and learning requests without changing the core modules.	Supports growth in users and request volume through scalable deployment.
3	Security & Data Privacy	Protect authentication data, user content and API credentials; keep secret keys outside client-side code.	No API keys exposed in frontend; authenticated access and protected environment variables.
4	Reliability & Availability	Handle invalid requests and temporary AI-service failures gracefully and return a clear user-facing message.	Graceful error handling with no application crash for expected failures.
5	Usability & Accessibility	Present answers, explanations, summaries, quizzes and recommendations in a simple student-friendly interface.	Core learning tasks can be completed with clear navigation and readable content.
6	Other	Keep the FastAPI modules maintainable and separate the Q&A, explanation, quiz, summary and learning-path functions.	Modular code structure with independently testable learning endpoints.